IV Fluid Maintenance Rates

Prodigy Anesthesia

IV Fluid Maintenance Rates

Calculating intravenous fluid maintenance rates can be an important component of the anesthetic management of adult patients undergoing complex surgical procedures and is a crucial component of pediatric anesthesia for any procedure. Intravenous fluid maintenance is generally calculated as an hourly rate for anesthesia purposes. To accommodate patients ranging in size from premature neonates to adults, a formula based on the patient's weight in kilograms is typically used that follows the form:

4 mL/kg/hr for the first ten kilograms of bodyweight

+

2 mL/kg/hr for the second ten kilograms of bodyweight

+

1 mL/kg/hr for each kilogram thereafter

This formula is not typically intuitive at first, so we'll use a few examples to demonstrate it.

What is the IV fluid maintenance rate for a 5 Kg infant?

Looking at our formula, it says that we should use 4 mL/hr for the first ten kilograms. This patient is only 5 Kg, so we simply multiply:

$$\frac{4\text{mL/hr}}{1\text{ Kg}} \times \frac{5\text{ Kg}}{1} = 20\text{ mL/hr}$$

So, a 5 Kg patient would have an IV fluid maintenance rate of 20 mL per hour. Let's try one a little more complicated:

What is the IV fluid maintenance rate for an 18 Kg child?

Looking at our formula, we should multiply 4 mL/hour X the first ten kilograms and 2 mL/hour X the second ten kilograms. Since our patient weighs 18 Kg, the formula should look like this:

$$\frac{4\text{mL/hr}}{1\text{ Kg}} \times \frac{10\text{ Kg}}{1} + \frac{2\text{mL/hr}}{1\text{ Kg}} \times \frac{8\text{ Kg}}{1} = 40 + 16 = 56\text{ mL/hr}$$

As you see, an 18 Kg patient would have (4 X 10) + (2 X 8) or 56 mL/hour for an hourly maintenance fluid rate. But what if the weight is over 20 kilograms? Let's look at one the hard way, and then we'll show you a shortcut.

What is the IV fluid maintenance rate for a 55 Kg woman?

Our formula accounts for the first 10 kilograms as 4 mL/kg/hour which equals 40 mL/hour plus 2 mL/Kg/hour for the second ten kilograms which adds another 20 mL/hour for a total of 60 mL/hour. That leaves an additional 35 kilograms that are added at 1 mL/kg/hour. So our formula looks like this:

$$\frac{4\text{mL/hr}}{1\text{ Kg}} \times \frac{10\text{ Kg}}{1} + \frac{2\text{mL/hr}}{1\text{ Kg}} \times \frac{10\text{ Kg}}{1} + \frac{1\text{mL/hr}}{1\text{ Kg}} \times \frac{35\text{ Kg}}{1} =$$

$$40\text{ml/hr} + 20\text{ mL/hr} + 35\text{ mL/hr} = 95\text{ mL/hr}$$

Now let's find the shortcut. What is the relationship between the patient's weight and their maintenance fluid rate? The weight is 55 Kg and the rate is 95 mL/hour. The maintenance rate is exactly 40 higher than the weight in kilograms.

Patients who weigh at least 20 Kg have an hourly IV maintenance rate that is their weight in kilograms plus 40.

So, a 30 Kg patient would have an hourly IV maintenance rate of 70 mL/hour, a 90 Kg patient would have a rate of 130 mL/hour, a 45 Kg patient would have a rate of 85 mL/hour, and so on. Note: this shortcut ONLY applies to patients who weigh at least 20 Kg. It does not apply to patients who weigh less than 20 Kg. Their weight must be calculated using the longer method. Remember, for some problems you may need to convert the weight into kilograms first.

IV Fluid Maintenance Rate Problems

1. What is the IV fluid maintenance rate for a 6 Kg patient?
2. What is the IV fluid maintenance rate for a 43 Kg patient?
3. What is the IV fluid maintenance rate for a 15 Kg patient?
4. What is the IV fluid maintenance rate for a 148 Kg patient?
5. What is the IV fluid maintenance rate for a 90 Kg patient?
6. What is the IV fluid maintenance rate for a 37 Kg patient?
7. What is the IV fluid maintenance rate for a 75 Kg patient?
8. What is the IV fluid maintenance rate for a 61 Kg patient?
9. What is the IV fluid maintenance rate for a 101 Kg patient?
10. What is the IV fluid maintenance rate for a 2 Kg patient?
11. What is the IV fluid maintenance rate for a 109 Kg patient?
12. What is the IV fluid maintenance rate for a 87 Kg patient?
13. What is the IV fluid maintenance rate for a 68 Kg patient?
14. What is the IV fluid maintenance rate for a 70 Kg patient?
15. What is the IV fluid maintenance rate for a 65 Kg patient?
16. What is the IV fluid maintenance rate for a 23 lb patient?
17. What is the IV fluid maintenance rate for a 115 lb patient?
18. What is the IV fluid maintenance rate for a 182 lb patient?
19. What is the IV fluid maintenance rate for a 104 lb patient?
20. What is the IV fluid maintenance rate for a 243 lb patient?
21. What is the IV fluid maintenance rate for a 145 lb patient?
22. What is the IV fluid maintenance rate for a 217 lb patient?
23. What is the IV fluid maintenance rate for a 158 lb patient?
24. What is the IV fluid maintenance rate for a 175 lb patient?
25. What is the IV fluid maintenance rate for a 120 lb patient?

IV Fluid Maintenance Rate Answers

1. 24 mL/hr
2. 83 mL/hr
3. 50 mL/hr
4. 188 mL/hr
5. 130 mL/hr
6. 77 mL/hr
7. 115 mL/hr
8. 101 mL/hr
9. 141 mL/hr
10. 8 mL/hr
11. 149 mL/hr
12. 127 mL/hr
13. 108 mL/hr
14. 110 mL/hr
15. 105 mL/hr
16. 41 mL/hr
17. 92 mL/hr
18. 123 mL/hr
19. 87 mL/hr
20. 151 mL/hr
21. 106 mL/hr
22. 139 mL/hr
23. 112 mL/hr
24. 120 mL/hr
25. 95 mL/hr